COMP4026 Computer Vision and Pattern Recognition

2nd Semester 2025-2026

Group Project

Title: Anonymised Facial Expression Recognition

Description

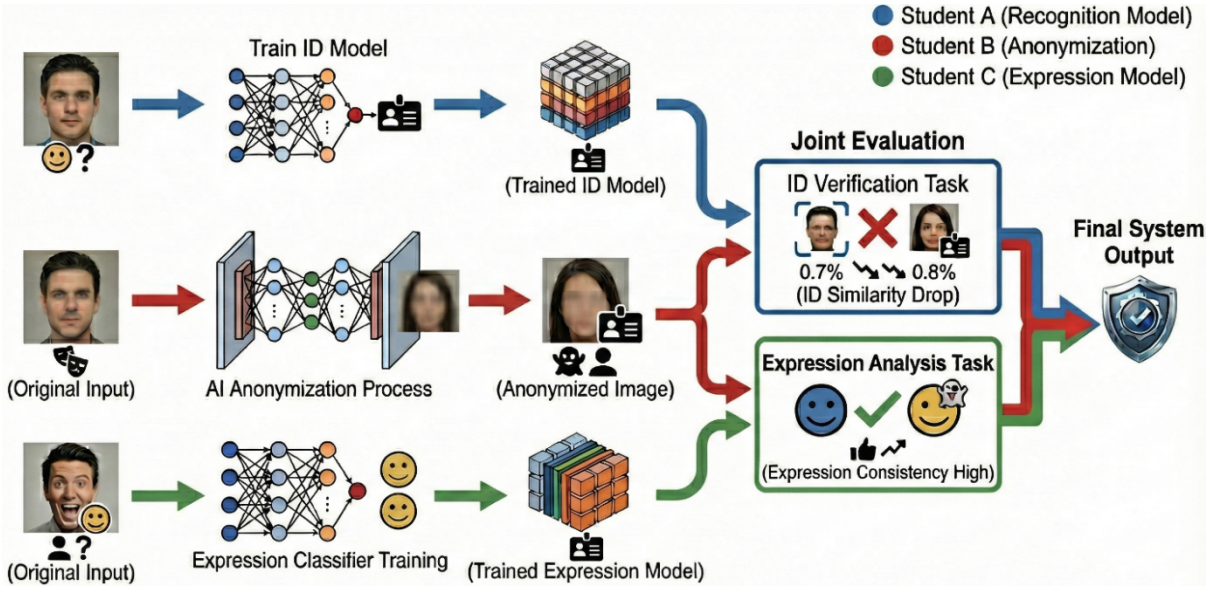
Many face-based applications, e.g. facial recognition and facial expression recognition, have been developed in the last decade. As facial images contain personal information, it is essential to protect individual privacy. In this project, students will study and work on a privacy-protected facial expression recognition project. The key challenge is maintaining facial expression recognition accuracy while the face images are privacy-protected (anonymised). The identity of the privacy-protected images should not be authenticated by a machine.

This project aims to build a prototype of an anonymised facial expression recognition application. To do this, the basic idea is to transform a face image into an anonymised version while preserving the facial expression. The anonymised face image is then used to recognise facial expressions.

A group of 3 students will work together on three tasks, namely face anonymisation, facial expression recognition and face recognition as shown in Figure 1. One suggestion is as follows.

- Student A: Develop a face recognition model.
- Student B: Develop a face anonymisation model.
- Student C: Develop a facial expression recognition model.
- Students A, B and C: Integration, experiments and evaluations

The goal is that the facial expressions in the original and anonymised face images are the preserved, while their identities are different.



Evaluation Metrics

- Face Recognition Metrics (Privacy)
  - **Accuracy** on original test set — validates the reliability of the ID classifier.
  - **Accuracy** on anonymised Images — evaluate the anonymisation strength.

- Expression Preservation Metrics (Utility)
  - **Expression Consistency Rate** — percentage of images with unchanged expression.

Project Evaluation Criteria

The following criteria will be used to evaluate your project:

- Understanding of the Task: Clear explanation of the purpose of the project.
- Methodology: Justifications for the tools, methods, and models used for both real and synthetic image training.
- Implementation and Comparison: Proper training, testing, and comparison of models.
- Efforts and Contributions: Demonstrated effort in generating synthetic data and analysing results.
- Presentation and Report Quality: Clarity and completeness of the final presentation and report.
- Teamwork

Dataset

Please use the following dataset for training on real images:

- Face Anonymization Model: CelebA-HQ
- Expression Classification: FER-2013
- Face Recognition Dataset: Pins Face Recognition

What and how to submit?

Each group must submit the following on Moodle.

- A project proposal with at most 1000 words, and 10 pages, including figures, tables and references. The proposal is in pdf format. Name your document with group number and last name of group members, e.g. Group1_Proposal_Wong_Chan_Lee.pdf
- A written final report in pdf format with at most 3000 words and 15 pages, including figures, tables and references. Same naming convention, i.e. Group1_Report_Wong_Chan_Lee.pdf
- Your presentation. Same naming convention, i.e. Group_1_Presentation_Wong_Chan_Lee.pptx
- Your source code. Same naming convention, i.e Group_1_Presentation_Wong_Chan_Lee.pptx

An overlength proposal/report will be subject to a penalty.

Final Report Format

While there is no standard structure of the report, you may consider the following structure.

1. **Title Page**: Include the project title, group number, and group member names.
2. **Abstract**: A concise summary of the project (maximum 300 words).
3. **Introduction**: Clearly explain the project objectives and background.
4. **Methodology**: Detail the techniques and tools used for real and synthetic image generation, training, and evaluation.
5. **Results and Discussion**: Present the performance metrics with a thorough analysis and comparison.
6. **Conclusion and Future Work**: Summarise the findings and suggest improvements or next steps.
7. **References**: Follow a standard citation style (e.g., IEEE).

Project Schedule

- **24 January 2026** *(5% deduction for late submission)*
  - Submit group member list on Moodle.
- **10 February 2026**
  - Project announcement.

- **17 March 2026***(20%)*
  - Submit a project proposal, including preliminary studies, methodology rationale, and initial design on Moodle.

- **20/21 April 2026** *(40%)*
  - 15-min presentation and demonstration. [All group members are required to present their responsible part(s)]

- **30 April 2026** *(40%)*
  - Submit final report with presentation slides and programme source code on Moodle.

Late submission policy: 10% deduction per day

Note:

- Declare the responsibility of each member during the presentation and in the report. Questions will be asked to specific member who is responsible for.
- Each member needs to present his/her part during the project presentation. Zero marks for no show.
- Final report needs to include a section "GenAI Usage Disclosure" and declare the use of third-party resources.